

First Lecture!

FPGA Technology Overview!

200b, 2nd floor teacher office

11 3th floor lab classes

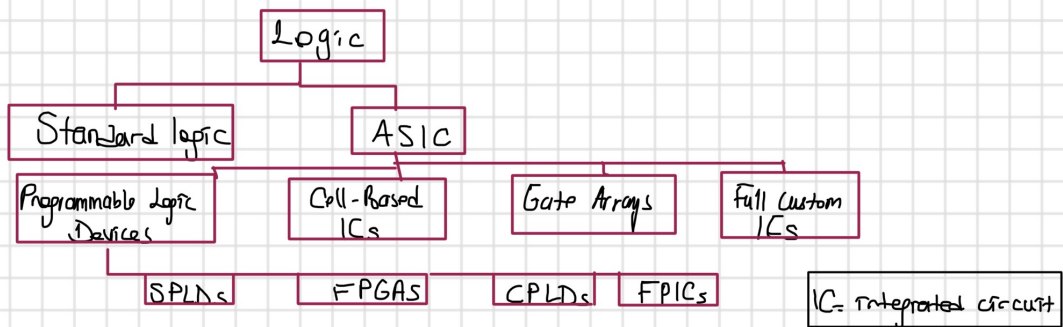
→ Wednesday 15.45 p.m. consultation hour

We don't have any exam but will ask basic lab exercises and individual projects.

~~VHDL~~ and Verilog

What is an FPGA?

FPGA → field programmable gate array



How do you make programmable circuit?

One time programmable

Fuses (destroy internal links with current)

Anti fuses (grow internal links)

PROM

Reprogrammable

EPROM

EEPROM

Flash

SRAM - volatile

} nonvolatile

} bunları silmek için ultra-violet ışığı gerekli

How do you configure an FPGA?

Creation a circuit design

VHDL

AHDL (Altera Hw Desc. L) for CPLD/FPGA

Verilog

System verilog

System C (HLS synthesis)

Compilation the HDL / C / SC code

Synthesis / implementation the design for the selected device

Download the compiled configuration

FPGAs offer many advantages over ASICs

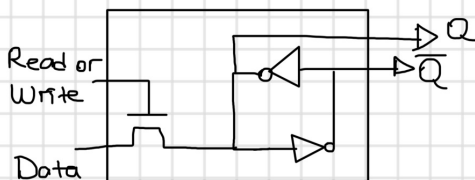
- Small development overhead
- No NRE (non-recurring engineering costs)
- Quick time to market
- No minimum quantity order
- Reprogrammable

What are the Resources of an FPGA

Basic components

- LUT (look-up-table)
- Flip-flops
- Multiplexers
- I/O blocks (programmable)
- Programmable switching matrices
- Interconnect
- Clocks

CMOS SRAM Cell



LUT table organization and creation

1	F1	0	0	0	0
1	F2	0	0	0	0
1	F3	0	0	0	0
1	F4	0	0	0	1

$$Q = F1 * F2 * F3 * F4$$

a) original 4-input AND function

LUT function

or LUT1: LUT3

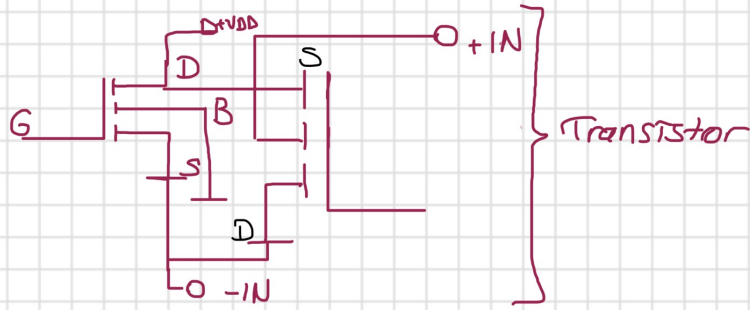
generic map (INIT => X"FE")
 port map (I0 => logic_result(1),
 I1 => arith_result(1),
 I2 => shift_result(1),
 O => qiu_group(1));

I2	I1	I0	O
000	00	0	0
000	00	1	1
000	01	0	1
000	01	1	1
000	10	0	1
000	10	1	1
000	11	0	1
000	11	1	1

2 Slice Virtex CLB

LE $\Rightarrow$ Flex 10K logic element

Switching Matrix



ISE 14.7

Xilinx.com